

EES UNIT 4 Question Bank

1. _____ pollutants are emitted directly from identifiable sources.

- a) Observable
- b) Tertiary
- c) Primary**
- d) Secondary

2. Which one of the following is NOT a primary pollutant ?

- a) nitrogen dioxides
- b) sulfuric acid**
- c) sulphur dioxide
- d) particulate matter
- e) carbon monoxide

3. The source of nearly half our pollution (by weight) is from the category.

- a) transportation**
- b) agriculture
- c) stationary source fuel consumption
- d) industrial processes
- e) none of the above

4. A pH of 4 is _____ times more acidic than a pH of 5.

- a) 5
- b) 10**
- c) 50
- d) 75
- e) 100

5. The vertical distance between Earth's surface and the height to which conventional movements extend is called the _____.

- a) vertical ratio
- b) stability layer
- c) conventional limiter
- d) mixing depth**
- e) turbulence factor

6. The major component of photochemical smog is _____.

- a) sulphuric acid
- b) ozone**
- c) particulate matter
- d) nitrogen dioxide
- e) carbon monoxide

7. Two of the most important atmospheric conditions affecting the dispersion of pollutants are the strength of the wind and the _____ of the air.

- a) stability**
- b) pressure
- c) visibility
- d) depth
- e) temperature

8. Which one of the following pollutants has shown the greatest percent decrease in concentration during the past years ?

- a) ozone
- b) sulfur dioxide
- c) nitrogen dioxide
- d) lead**
- e) particulate matter

9. This colorless, odorless, poisonous gas is the most abundant primary pollutant.

- a) carbon dioxide
- b) nitrogen dioxide
- c) sulfur dioxide
- d) methane
- e) none of the above**

10. Noise pollution is best described as _____.

- a) barking dogs
- b) any unwelcome sound**
- c) singing groups
- d) rock music

11. Decibel (dB) is the measure of _____.

- a) sound intensity**
- b) decorated bells
- c) sound language
- d) sound frequency

12. Sound intensity is the air pressure caused by _____.

- a) light waves
- b) sound waves**
- c) space waves
- d) transverse waves

13. Sound intensity measured by decibels is not done by addition but rather through _____.

- a) Division
- b) Subtraction
- c) Multiplication**
- d) Addition

14. Painful sensation in the ear is usually at the level of _____.

- a) 1000 to 1200 dBA
- b) 0 to 20 dBA
- c) 120 to 130 dBA**
- d) 50 to 70 dBA

15. The dangerous decibels of noise _____.

- a) can be altered by distance**
- b) are not altered by how close you are standing to speakers
- c) cannot be altered by distance
- d) can be stopped altogether by trees

16. It is important to distinguish between point source and non point source pollution because point source pollution is much easier to control. Which of the following is an example of non point source pollution ?

- a) domestic sewage
- b) storm-water runoff from a storm drainpipe
- c) discharge pipe from a meatpacking point
- d) agricultural fertilizers from cropland**

17. In which of the following locations is thermal pollution the biggest problem ?

- a) Surface water**
- b) Air
- c) Soil
- d) Groundwater

18. Primary sewage treatment removes what percentage of the biochemical oxygen demand (BOD) from domestic sewage ?

- a) 3
- b) 60
- c) 90
- d) 33**

19. Some of the oil pollution of our oceans has always occurred naturally through natural leaks in the ocean floor. However, this source accounts for only about 9% of the oil that reaches our oceans.

Which of the following is the primary source of pollution in the oceans ?

- a) Harbor pollution
- b) Inland sources, such as motor vehicle crankcases**
- c) Oil tanker accidents
- d) None of these

20. Which of the following pathogenic microorganisms is unlikely to be contracted through water ?

- a) Amoebic dysentery
- b) Typhoid fever
- c) Hepatitis
- d) Cholera
- e) West Nile virus fever**

21. At what stage in the wastewater treatment procedure would nutrients such as nitrogen and phosphorus be removed in a secondary treatment facility ?

- a) On the trickling filter
- b) At the bar screen
- c) Never**
- d) In settling tank

22. What pollutant has only recently (in the past 10 years) been discovered in many inland waters of the developed nations ?

- a) Artificial pesticides
- b) Kerosene derivatives
- c) Hormone disruptors**
- d) Nitrates

23. Which of the following is directly or indirectly related to essentially all of our environmental problems ?

- a) Depletion of resources
- b) Population growth**
- c) Technology
- d) Pollution
- e) Destruction of rainforests

24. The environmental cause of increase in skin cancer cataract and mutation is _____.

- a) Acid rain
- b) Global warming
- c) Nuclear winter
- d) Ozone depletion**

25. Greenhouse effect leading to the warming up of the atmosphere is due to _____.

- a) Infrared rays reaching to Earth
- b) UV rays reaching Earth
- c) Increase in temperature due to increase in CO₂ level**
- d) Depletion of Ozone layer

26. Acid rain _____.

- a) increases the pH of soil
- b) decreases the pH of soil**
- c) does not change the pH of soil
- d) increases the fertility of soil

27. Which one is not greenhouse gas ?

- a) CO₂
- b) Ozone**
- c) CH₄
- d) CFC

28. Our environment is presently not degrading due to _____.

- a) deforestation
- b) overgrazing
- c) rapid industrialization
- d) urbanization
- e) none of the above**

29. Which one of the following is NOT included in Earth's climate system ?

- a) Atmosphere
- b) Solid Earth
- c) Biosphere
- d) Hydrosphere
- e) Solar sphere**

30. Which one of the following is NOT a significant natural cause of climate change ?

- a) Solar variability
- b) Variations in Earth's orbit
- c) Volcanic activity
- d) Plate tectonics

e) Erosion of the land

31. The combustion of coal and oil is a major source of which greenhouse gas ?

- a) Ozone
- b) Nitrous oxide
- c) Methane
- d) CFCs

e) None of the above

32. Which of the following is associated with the astronomical theory of climate change ?

- a) Eccentricity of Earth's orbit
- b) Obliquity of Earth's axis
- c) Precession of Earth's axis

d) All of the above

e) Both precession and obliquity of Earth's axis

33. Which of the following is a significant contributor to increased levels of atmospheric carbon dioxide ?

a) Deforestation

- b) Aerosol spray can gases
- c) Refrigerant leakage
- d) Large rice paddies
- e) None of the above

34. The slow wobbling of Earth as it spins on its axis is called _____.

- a) eccentricity
- b) obliquity
- c) aphelion
- d) precession**
- e) extension

35. Which of the following is a possible consequence of a greenhouse warming ?

- a) A change in agricultural productivity
- b) A rise in sea level
- c) New weather patterns
- d) Alteration of the world's water resources
- e) All of the above**

36. Oxygen isotope analysis measures the _____.

- a) Amount of radioactive oxygen in the air
- b) Oxygen content of ocean water
- c) Ratio of two different types of oxygen atoms**
- d) Oxygen content of organic sediment
- e) rate of oxygen depletion

37. Which one of the following is NOT a greenhouse gas ?

- a) Nitrous oxide
- b) Methane
- c) Carbon dioxide
- d) Oxygen**
- e) All are greenhouse gases

38. When determining climate change, oxygen isotope measurements are made from an analysis of _____.

- a) ocean water
- b) lava
- c) cores from old trees
- d) glacial ice**
- e) none of the above

39. Along with carbon dioxide, _____ is largely responsible for the greenhouse effect of the atmosphere.

- a) nitrogen
- b) hydrogen
- c) sulphur dioxide
- d) oxygen
- e) water vapour**

40. Which of the following help scientists discover how climate has changed over broad expanses of time ?

- a) Sea floor sediments
- b) Oxygen isotope analysis
- c) Paleosols (buried soil)
- d) All of the above**
- e) Only sea-floor sediments and oxygen isotope analysis

41. The greenhouse effect is caused by _____.

- a) Greenhouse gases in the lower atmosphere absorbing solar radiation
- b) Greenhouse gases in the lower atmosphere absorbing radiation from the Earth's surface, and preventing much of it escaping into space**
- c) Too much heat in the atmosphere
- d) Too much sunshine reaching Earth

42. Which of these natural events affect the climate ?

- a) Thawing permafrost
- b) The path of the Earth around the sun
- c) Volcanic eruptions
- d) All of the above**

43. Which of these greenhouse gases is most abundant in the atmosphere ?

- a) Carbon dioxide
- b) Methane
- c) Nitrous oxide
- d) Water vapour**

44. What greenhouse gas was not present in the atmosphere in pre-industrial times (before 1750) ?

- a) Trichlorofluoromethane**
- b) Carbon dioxide
- c) Nitrous oxide
- d) Methane

45. Which of these countries has the highest per capita carbon dioxide emissions ?

- a) United States
- b) Australia
- c) Saudi Arabia**
- d) China

46. Which of the following activities contributes the most to carbon emissions globally ?

- a) Agriculture
- b) Transport
- c) Forestry
- d) Energy supply**

47. The 1987 Montreal Protocol was signed for which of the following reasons ?

- a) To phase out the use of CFC's, found to be causing depletion of the ozone layer**
- b) To begin converting from fossil fuel use to more renewable energy sources to reduce the anthropogenic greenhouse effect
- c) To stop the global trade in products made from endangered tigers
- d) To ban nuclear testing in tropical oceans

48. One effect of global warming is a rise in sea levels, which certainly could affect coastal cities that are little above current sea level. How much has the sea level risen in the past 100 years ?

- a) 1-2 inches
- b) 6-8 feet
- c) 6-8 inches**
- d) 1-2 feet

49. What would the Kyoto Protocol do ?

- a) It would not allow developing countries like China to modify their coal-burning equipment to prevent further increases in carbon dioxide levels
- b) It would curb all CO₂ emissions from all countries to bring them about 15 to 20% down from the levels reached in 1990
- c) It would completely ban production of CFC's
- d) By 2012, it would curb CO₂ emissions from all developed countries to bring them about 6 to 8% down from the levels reached in 1990**

50. Which statement below is correct ?

- a) Normal rain's pH varies by more than 3 pH units
- b) Normal rain has perfectly neutral reaction (pH=7)
- c) Normal rain is mildly acidic because of the carbonic acid (pH < 7)**
- d) It is impossible to tell
- e) Normal rain is mildly alkaline (pH > 7)

51. Which statement best describes the current situation with the ozone hole ?

- a) The ozone hole has grown about 50% larger
- b) The ozone hole has shown a major decline after the ozone-depleting chemicals were banned under the Montreal Protocol
- c) The ozone hole remained more or less constant in size since 2000, despite a slight reduction in the amount of CFC's in the stratosphere**
- d) Ozone holes of about equal size are now developing in both the Arctic and Antarctica

52. Which strategy below will not assist us in preventing acid rain ?

- a) Installing scrubbers at power plants
- b) Switching to wind power as a source of electricity
- c) Using used rubber tires on children's playgrounds**
- d) Switching to low-sulphur coal

53. The pH scale is logarithmic; that is, each unit on it is 10 times greater than the previous one.

- a) True**
- b) False

54. Even with a complete CFC ban worldwide, it will take the ozone layer 50 to 100 years to recover.

- a) True**
- b) False

55. Ozone is a pollutant at ground level, but is a desirable chemical in the stratosphere because it blocks dangerous UV radiation.

- a) True**
- b) False

56. An ozone hole can occur only over Antarctica.

- a) True
- b) False**

57. The projected effects of global warming are all negative.

- a) True
- b) False**

58. Global warming over the 21st century will likely result in a 10 meter rise in ocean level.

- a) True
- b) False**

59. Because warming is global, Earth is warming up uniformly everywhere.

- a) True
- b) False**

60. Reductions in methane emissions could help mitigate global warming.

- a) True**
- b) False

61. A global circulation models predict a future that is warmer than today for the next 50-100 years.

- a) True**
- b) False

62. In the past century, the amount of precipitation worldwide has decreased.

- a) True
- b) False**

63. To stop global warming by the end of the 21st century, we would need to reduce greenhouse gas emissions much more than what's required under Kyoto Protocol (6 to 8%), perhaps by as much as 80% within the next 20-50 years.

- a) True**
- b) False

Additional Questions

1. Natural pollution is caused by _____.

- a) volcanic eruptions**
- b) vehicle emissions
- c) cutting of woods
- d) smoke from industry

2. DDT is the _____.

- a) Natural pollutants
- b) Primary pollutants**
- c) Secondary pollutants
- d) None of these

3. Peroxy-acetyl nitrate (PAN) is formed by the reaction of two primary pollutant, namely nitrogen oxide and _____ released from motor vehicle in presence of sunlight.

- a) hydro-neutrons
- b) hydro-carbons**
- c) water-carbons
- d) SO₂

4. The pollutants which occur in nature but become pollutant when their concentration reaches beyond a threshold value in the environment are called as _____.

- a) Primary pollutants
- b) Quantitative pollutants**
- c) Secondary pollutants
- d) Qualitative pollutants

5. The pollutants which do not occur in the environment but are passed into it through human activity are called as _____.

- a) Qualitative pollutants**
- b) Quantitative pollutants
- c) Natural pollutants
- d) None of these

6. DDT is an example of _____.

- a) Qualitative pollutant**
- b) Quantitative pollutant
- c) Natural pollutant
- d) None of these

7. CO₂ is an example of _____.

- a) Artificial pollutant
- b) Quantitative pollutant**
- c) Qualitative pollutant
- d) None of these

8. The pollutants which are waste products and they are slowly degraded by microbial action are called as _____.

- a) Biodegradable pollutants**
- b) Non-biodegradable pollutants
- c) Artificial pollutants
- d) None of these

9. In terms of origin, pollution may be natural and _____.

- a) super natural
- b) sub-natural
- c) anthropogenic**
- d) anti-anthropogenic

10. The man made pollution, such as industrial pollution, agricultural pollution etc. are called as _____.

- a) Anthropogenic**
- b) Anti-anthropogenic
- c) Semi-anthropogenic
- d) Natural anthropogenic

11. On the basis of physical state, air pollutants are of two types, one is gaseous and other is _____.

- a) particulate**
- b) semi-particulate
- c) tri particulate
- d) fog

12. The pollutants occur as solid and liquid state are referred as _____.

- a) Foam
- b) Fog
- c) Particulate pollutant**
- d) Gaseous pollutant

13. The vapour of compounds whose boiling point is below 200°C are included in _____.

- a) particulate
- b) gaseous pollutant**
- c) smog
- d) mist

14. The smaller particles such as dust, smoke etc. which remain suspended for long period in air are called _____.

- a) Setteable particles
- b) Suspended particles**
- c) Gaseous particles
- d) Micro pollutants

15. The particles larger than 10 μm in diameter are called _____.

a) Gaseous pollutants

b) Setteable particles

c) Non-setteable particles

d) Suspended particles

16. The larger suspended particles with more than 1 μm in diameter are called _____.

a) Dust and Mist

b) Gaseous and semi-gaseous pollutants

c) Fog and Frost

d) Aqua and Rox-mixed

17. Cotton dust produce lung fibros is, also called as _____.

a) byssinosis

b) silicosis

c) siderosis

d) none of these

18. Air bonne organic materials such as spores, pollen, bacteria, fungi cause several diseases and allergic reactions called as _____.

a) hay fever

b) T.B.

c) Ronbix

d) Altra-rix

19. BOD stands for _____.

a) Biological Oxygen Demand

b) Bio-Chemical Oxygen Demand

c) Bio-Chemical Ozone Demand

d) Biological Oxygen Difficiency

20. BOD expressed in _____.

a) milligrams of carbon per litre of water

b) milligrams of sulphur per litre of water

c) milligrams of oxygen per litre of water

d) microorganisms of oxygen per litre of water

21. COD stands for _____.

- a) Carbon Oxygen Demand
- b) Carbon Obtained Demand
- c) Chemical Carbon Demand
- d) Chemical Oxygen Demand**

22. An accidental discharge of petroleum in oceans is called _____.

- a) oil leaks
- b) oil milk
- c) oil spills**
- d) oil waste

23. Water having DO content below _____ is considered to be highly polluted.

- a) 4.0 mg/L**
- b) 8.0 mg/L
- c) 11.2 mg/L
- d) 0.01 mg/L

24. The species survive in highly polluted water with low DO content are recognized as _____ for polluted water.

- a) Re-survive
- b) Top species
- c) Indicator species**
- d) Gama species

25. Water contaminated with cadmium can cause _____ disease.

- a) Itai Itai disease**
- b) La la disease
- c) Minamata disease
- d) Sasu disease

26. Pesticides are also called as _____.

- a) Supercides
- b) biocides**
- c) nano-cides
- d) catcides

27. Increase in the concentration of soluble salts in the soil is called as _____.

- a) soilification
- b) soluble-soil
- c) salination**
- d) desalination

28. Radioactivity is a property of certain elements to spontaneously emit protons. Protons means emission of _____.

- a) alpha particles**
- b) beta particles
- c) gamma particles
- d) none of these

29. The unit of sound level is _____.

- a) decibel**
- b) noise-meter
- c) dolphan
- d) dub

30. Noise level beyond _____ dB cause physical discomfort.

- a) 1120
- b) 10
- c) 120**
- d) 0

31. Environmental pollution means adding _____ to environment.

- a) impurity**
- b) chemicals
- c) metals
- d) non-metals

32. Industrial pollution is _____.

- a) Man made pollution**
- b) Natural pollution
- c) Self pollution
- d) None of these

33. Volcanic eruptions release gases and volcanic _____.

- a) Ash**
- b) Chemical
- c) Dater
- d) Carbon

34. Decay of organic matter in marshy places releases _____.

- a) Marsh gaps**
- b) Chemical gas
- c) Wood sap
- d) None of these

35. _____ is a chemical substance found within an organism.

- a) Xenobiotic**
- b) Noxenobiotic
- c) Virus
- d) Fungi

36. Chronic exposure to benzene at sufficient concentrations is known to be associated with higher incidence of _____.

- a) leukemia**
- b) cough
- c) hair loss
- d) eye irritation

37. The total mass of the earth's hydrosphere is about _____ 18 tonnes.

- a) 1
- b) 2
- c) 3
- d) 1.4**

38. The total mass of Earth's hydrosphere is about of the Earth's total mass.

- a) 1
- b) 2
- c) 3
- d) 0.023%**

39. Surface water is _____ % of the fresh water.

- a) 1
- b) 0.1
- c) 0.3**
- d) 10

40. Sub surface water or ground water is approximately _____ % of fresh water.

- a) 10.1
- b) 20.1
- c) 30.1**
- d) 40.1

41. Sub surface water or ground water is about _____% of the total water available on Earth.

- a) 1**
- b) 2
- c) 3
- d) 10

42. The fresh water located in the pore space of soil and rocks is called as _____.

- a) sub surface water**
- b) well water
- c) ice water
- d) polluted water

43. _____% of fresh water is entrapped in icecaps and glaciers.

- a) 68.7**
- b) 11%
- c) 1.1
- d) 11.8

44. The Himalayas, which are often called _____.

- a) The roof of the world**
- b) The cap of the world
- c) The acme of the world
- d) Glory of the world

45. Factories are the _____ of water pollution.

- a) point source**
- b) non-point source
- c) artificial source
- d) none of these

46. _____ are the source of pollutants that cannot be traced to a single point of discharge.

- a) Point source
- b) Non-point source**
- c) Artificial source
- d) None of these

47. _____ is caused due to poor drinking water.

- a) Cholera**
- b) Malaria
- c) Dengue
- d) Asthma

48. As per BIS, pH value of drinking water is _____.

- a) 1-2
- b) 2-4
- c) 4-8
- d) 6.5-8.5**

49. As per BIS, the acceptable limit of dissolved solids (m/l) in drinking water is _____.

- a) 100
- b) 200
- c) 500**
- d) 1000

50. In treatment plant, the process of removal of suspended solids trapped in floc is called as _____.

- a) coagulation
- b) pre-chlorination
- c) chlorination
- d) sedimentation**

51. The killing of bacteria means _____.

- a) Disinfection**
- b) Sedimentation
- c) Aeration
- d) Filtration

52. _____ is the water draining from your house's sinks, bathtubs, laundry machine which can be used to water plants.

- a) Yellow water
- b) Grey water**
- c) Black water
- d) Blue water

53. The water from toilets, which needs to be treated is called as _____.

- a) Black water**
- b) Blue water
- c) Yellow water
- d) Brown water

54. The lawn made by drought-resistant grasses is called as _____.

- a) ECO lawn**
- b) Show lawn
- c) Biotic lawn
- d) Shower lawn

55. Domestic sewage is _____% of pure water.

- a) 99.9**
- b) 9.9
- c) 90.9
- d) 100

56. In domestic sewage the percent of pollutants is _____.

- a) 0.1%**
- b) 0.001%
- c) 99%
- d) 100%

57. VOC stands for _____.

- a) Volcanic Organic Compound
- b) Volatile Organic Compound**
- c) Various Organic Compound
- d) Volcanic Obtain Compound

58. 1, 3-butadiene is dangerous compound which is often associated with _____ users.

- a) industrial**
- b) non-industrial
- c) agricultural
- d) none of these

59. The word 'smog' is coined out of smoke and _____.

- a) smell
- b) fog**
- c) flock
- d) foc

60. POP stands for _____.

- a) Persistent Organic Pollutants**
- b) Proper Organic Pollutants
- c) Properly Organised Pollutants
- d) Purposely Organised Pollutants

61. Radon gas from radioactive decay within Earth's _____.

- a) crust**
- b) surface
- c) water
- d) air

62. Radon gas is formed from decay of _____.

- a) Radium**
- b) Methane
- c) Sodium
- d) Atom

63. Radon gas is responsible for _____.

- a) bone cancer
- b) lung cancer**
- c) skin cancer
- d) blood cancer

64. The worst short term civilian pollution crisis in India was the 1984 _____ disaster.

- a) Nagpur
- b) Delhi
- c) Bhopal**
- d) Kanpur

65. India where buses now use compressed natural gas to help eliminate the _____ smog.

- a) pea-soup**
- b) pea-click
- c) pea-rock
- d) pea-pea

66. _____ is the chemical reaction of sunlight, nitrogen oxides and volatile organic compounds in the atmosphere.

- a) Photochemical smog**
- b) Persistent smog
- c) Permanent smog
- d) None of these

67. An electrostatic precipitator (ESP) is a _____ device.

- a) filtration**
- b) sedimentation
- c) coagulation
- d) none of these

68. As per national ambient air quality standards, the amount of ozone (O₃) in µg/m³ (1 hour) in residential area is _____.

- a) 200
- b) 300
- c) 180**
- d) 90

69. _____ are efficient at removing dust and particles.

- a) venture scrubbers**
- b) ventilators
- c) high volume sampler
- d) none of these

70. As per national ambient air quality standards the amount of carbon monoxide (CO) in mg/m³ in 1 hour is _____.

- a) 01
- b) 100
- c) 04**
- d) 11

71. Noise pollution is measured in _____.

- a) Decibel unit**
- b) Cobalt unit
- c) Sound unit
- d) Hearing unit

72. Noise stands for residential area in day time is _____ dB (A).

- a) 10
- b) 20
- c) 25
- d) 55**

73. Noise standards in silence zone in a day time is _____ dB (A).

- a) 10
- b) 20
- c) 30
- d) 50**

74. Silence zone is referred as area upto _____ meters around such as hospital, educational institutions and courts.

- a) 100**
- b) 200
- c) 300
- d) 400

75. WEEE stands for _____.

- a) Waste Electrical and Electronic Equipment**
- b) Waste Environmental and Electronic Equipment
- c) Worst Electrical and Electronic Equipment
- d) None of these

76. Organic matter constitute _____% of the municipal solid waste generated in India.

- a) 10-20
- b) 35-40
- c) 40-60**
- d) 80-90

77. CPCB stands for _____.

- a) Central Pollution Control Board**
- b) Centre for Pollution Conserve Board
- c) Central Protection Control Board
- d) Central Protection Conserve Board

78. SPCB stands for _____.

- a) State Protection Control Board
- b) State Pollution Control Board**
- c) Standard Pollution Control Board
- d) State Pollution Conserve Board